

**PROJECT**

DEPARTMENTAL STORE
MANAGEMENT SYSTEM

NAME

- ZAINAB SARFRAZ
- ADEEN TARIQ
- EMAN QASIM

REG NO

- 24-NTU-CS-FL-1289
- 24-NTU-CS-FL-1242
- 24-NTU-CS-FL-1255

SEMESTER

3rd

COURSE CODE/TITLE

CSC2073/SOFTWARE ENGINEERING
AND FUNDAMENTALS

SECTION/DEPARTMENT

BSAI/CS

SUBMITTED TO

DR. MUHAMMAD ADEEL

SUBMISSION DATE

30th December, 2025

DEPARTMENTAL STORE MANAGEMENT SYSTEM

PROJECT OVERVIEW:

The Departmental Store Management System is a software application designed to automate and streamline the day-to-day operations of a medium to large trade or departmental store. It replaces traditional manual record-keeping with a fast, accurate, and user-friendly digital system.

OBJECTIVES:

- To manage store items efficiently
- To generate automatic bills
- To track inventory and stock levels
- To prevent over-selling
- To provide daily/monthly sales reports
- To maintain customer and supplier information
- To reduce manual errors and save time

STEPS IN USE CASE MODELLING:

There are the following steps of use case modelling:

Step 1: Identify and Define the Actors:

This first step is about identifying everyone and everything that interacts with your system.

- Customer
- Cashier
- Admin
- Inventory Manager

Step 2: Identify and Define the Use Cases:

This step involves determining the specific goal-oriented tasks that each actor performs using the system.

➤ **Customer Use Cases:**

These are the primary actions a shopper performs.

- Search Products
- View Product Details
- Add Product to Cart
- Remove Product from Cart

- Update Product in Cart
- View Cart
- Place order

➤ **Cashier Use Case:**

These involve handling the transactions at the checkout.

- Print Receipt
- Process Payment
- Exchange
- Apply Discount

➤ **Admin Use Case:**

These are the core maintenance and security tasks.

- Configure System Setting
- Manage User Accounts

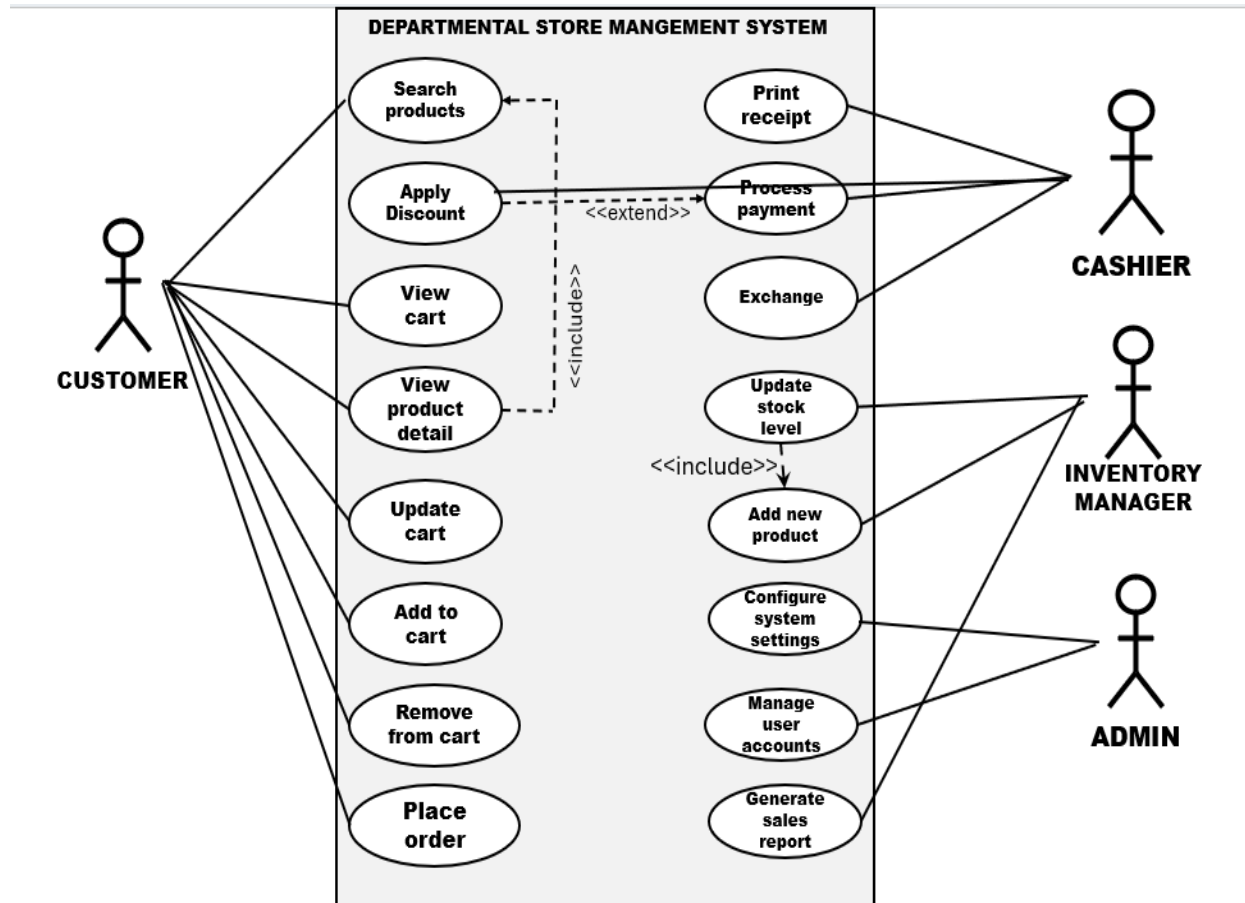
➤ **Inventory Manager Use Case:**

These focus on managing the actual items sold in the store.

- Add new product
- Update Stock Level
- Generate Sales Report

Step 3: Construct use case model diagram:

This step is about creating the **visual blueprint** of your system's functionality and scope.



Step 4: Documents Business Requirements Use-case

Narratives:

This step moves from the visual model to the detailed text documentation of each function. These narratives capture the how and why of the system's behaviour.

ELABORATED USE CASE SPECIFICATIONS:

➤ **Search Product:**

- **Implementation priority:** high
- **Actors:** Customer
- **Summary:** The customer submits a query (text, barcode, or category selection) to find and display a list of relevant products from the store inventory.
- **Preconditions:** The system is running, and the product catalogue database is accessible.
- **Post conditions:** A results page is displayed showing all matching products and their current pricing/availability.
- **Extends:** None
- **Uses:** None.
- **Normal course of events:**
 - **Actor:** Customer enters search term (e.g., "milk").
 - **System:** System validates the input and queries the product database.
 - **System:** System retrieves a list of products matching "milk."
 - **System:** System displays the search results page to the customer.
 - **Actor:** Customer views the results.
 - **System:** Use Case Ends.
- **Alternative path:**
 - **Path 1:** The customer selects a category link (e.g., "Dairy") instead of entering text. The system retrieves all products under that category.
 - **Path 2:** The search query returns to zero matches. The system displays a "No Results Found" message but still ends successfully.
- **Exception:** If the system is unable to connect to the product database in step 2, the system must display an error message (e.g., "System Error: Unable to retrieve data") and log into the failure.

- **Assumption:** It is assumed that the product database records are structured correctly and contain valid product IDs, names, and pricing data.

➤ **Add item to cart:**

- **Implementation priority:** High
- **Actors:** Customer
- **Summary:** The user selects a product and specifies a quantity to be added to the current shopping cart or transaction session.
- **Preconditions:** The system must be displaying a searchable list of products. The product ID must be valid and recognized by the system.
- **Post conditions:** The item is successfully added to the Cart/Transaction object. The current Cart Subtotal and item count are updated.
- **Extends:** None
- **Uses:** None.
- **Normal course of events:**
 - **Actor:** User selects a product ID (e.g., clicks 'Add to Cart' or Cashier scans the barcode).
 - **System:** System validates the product ID and quantity (Uses **Validate Stock Availability**).
 - **System:** System retrieves the unit price (Uses **Retrieve Product Price/Details**).
 - **System:** System adds the item and quantity to the Cart/Transaction object.
 - **System:** System updates and displays the new cart subtotal.
 - **System:** Use Case Ends.
- **Alternative path:**
 - **Path 1:** If the requested quantity exceeds the available stock, the system displays an alert ("Only 5 available") and restricts the quantity added to the stock limit. Use Case ends successfully with a partial addition.
 - **Path 2:** If the item already exists in the cart, the system simply increases the quantity count for that existing item line instead of adding a new line.
- **Exception:** The system fails to write the new cart details to the database, the system must display an "Order System Unavailable" message and roll back the transaction state.

- **Assumption:** It is assumed that the Product table contains valid and non-null values for both ProductID and UnitPrice.

SRS DOCUMENT ELEMENTS:

- **Purpose of software being developed:**
 - The purpose of this document is to describe the complete software requirements for the Departmental Store Management System (DSMS). This SRS is intended to provide a clear understanding of the system for developers, instructors, and stakeholders.
- **Overall description of the software:**
 - **Scope:** The Departmental Store Management System is a software application designed to automate the daily operations of a department store. The system will help manage products, inventory, sales, billing, and reporting. It aims to replace manual record-keeping and improve efficiency and accuracy.
 - **Product Perspective:** The Departmental Store Management System is a standalone desktop-based application integrated with a database. It is designed specifically for managing department store operations.
- **Functionality of the software:**
 - **Product function:** The major functions of the system include:
 - User authentication (login and logout)
 - Product management
 - Category management
 - Inventory control
 - Sales and billing processing
 - Report generation
 - **System features:**
 - **Login System:**
 - ✦ **Description:** The system allows authorized users to log in using valid credentials.
 - ✦ **Functional Requirements:**
 - ◆ The system shall allow users to enter a username and password.
 - ◆ The system shall authenticate user credentials.
 - ◆ The system shall display an error message for invalid login attempts.
 - **Product Management:**
 - ✦ **Description:** The admin can manage product records in the system.
 - ✦ **Functional requirements:**

- ♦ The system shall allow adding new products.
- ♦ The system shall allow updating product details.
- ♦ The system shall allow deleting products.
- ♦ The system shall display a list of all products.

➤ **Category Management:**

✦ **Description:** The system allows products to be organized into categories.

✦ **Functional requirements:**

- ♦ The system shall allow adding new categories.
- ♦ The system shall allow updating category details.
- ♦ The system shall allow deleting categories.

➤ **Inventory Management:**

✦ **Description:** The system automatically manages stock levels.

✦ **Functional requirements:**

- ♦ The system shall update product stock quantities.
- ♦ The system shall generate low-stock alerts.
- ♦ The system shall generate inventory reports.

➤ **Sales and Billing:**

✦ **Description:** The cashier processes customer purchases and generates bills.

✦ **Functional requirements:**

- ♦ The system shall allow product selection for sales.
- ♦ The system shall calculate the total bill automatically.
- ♦ The system shall generate and print receipts.

➤ **Reports:**

✦ **Description:** The system generates reports for management purposes.

✦ **Functional requirements:**

- ♦ The system shall generate daily sales reports.
- ♦ The system shall generate monthly sales reports.
- ♦ The system shall generate inventory reports.

○ **Performance of software in a production situation:**

• **Performance Requirements:**

- The system shall provide fast response times.
- The system shall efficiently handle large numbers of records.

○ **Non-functional requirements:**

• **Performance requirements:**

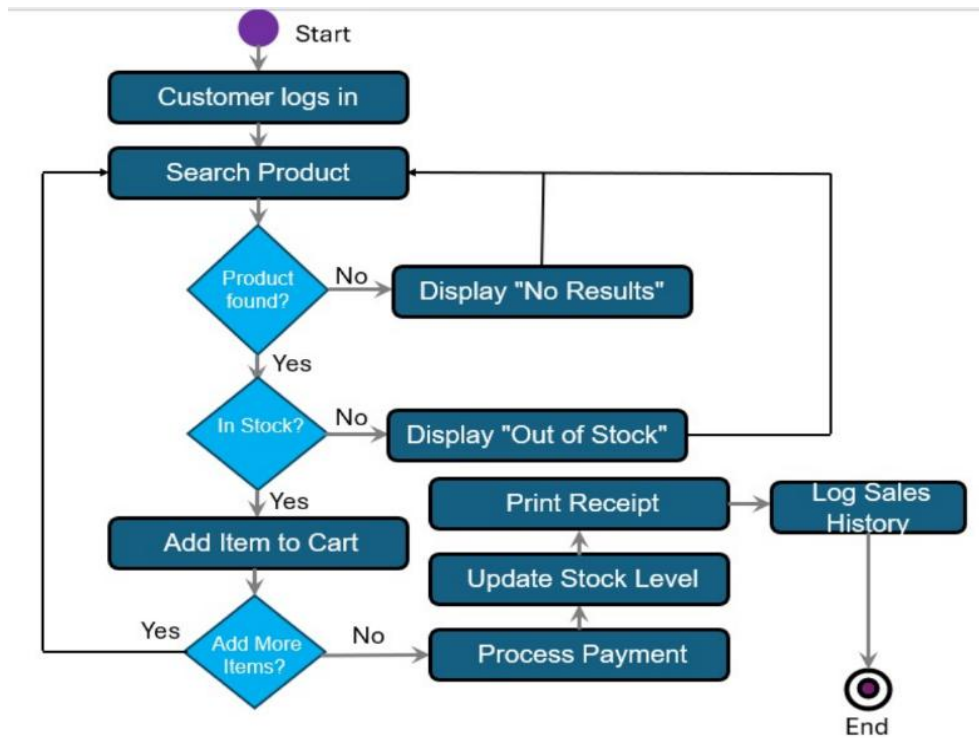
- The system shall provide fast response times.

- The system shall efficiently handle large numbers of records.
- **Security requirements:**
 - The system shall provide password-protected access.
 - Unauthorized users shall not be allowed access.
- **Reliability requirements:**
 - The system shall operate without frequent failures.
 - Data loss shall be prevented.
- **Usability requirements:**
 - The system shall be easy to use.
 - Minimal training shall be required for users.
- **External interfaces:**
 - **User Interface:**
 - Graphical and user-friendly interface.
 - Form-based navigation.
 - **Hardware Interface:**
 - Keyboard
 - Mouse
 - Printer for receipt generation
 - **Software Interface:**
 - Database Management System
 - Operating system services
- **Design Constraints:**
 - **Design and Implementation Constraints:**
 - The system will be desktop-based.
 - The system will run on Windows operating systems.

DIAGRAMS:

ACTIVITY DIAGRAM:

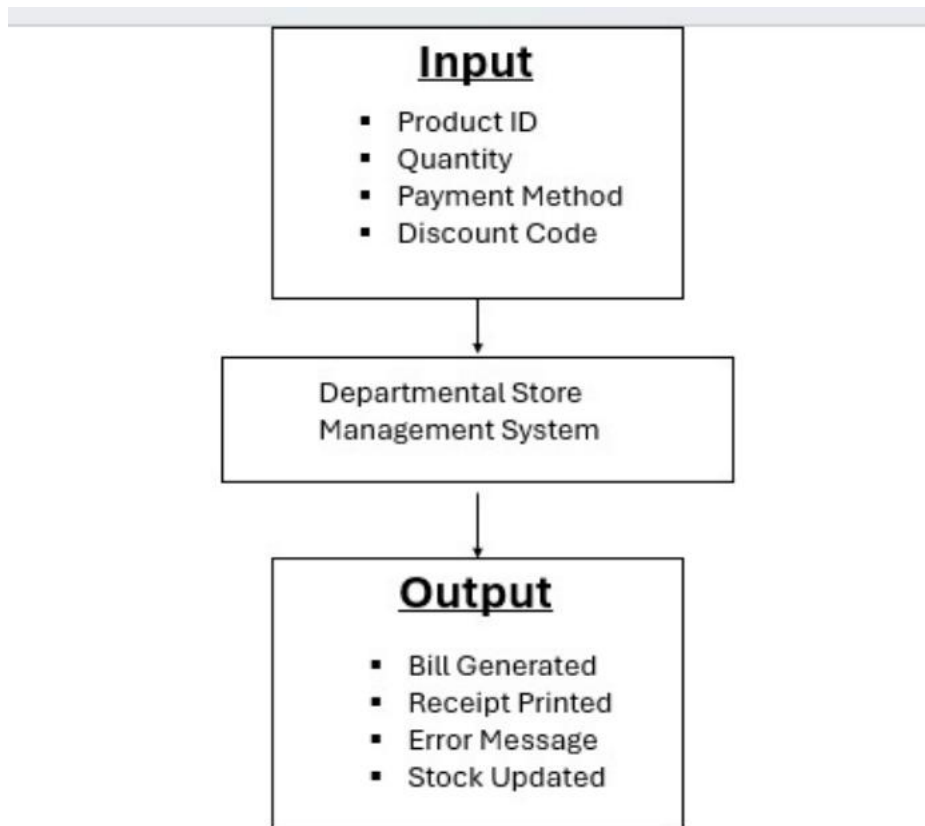
The activity diagram of the Departmental Store Management System illustrates the step-by-step flow of activities performed by different users during system operation. It starts when the customer searches and selects products, adds items to the cart, and reviews the cart details. After placing the order, the cashier processes the payment by applying discounts if applicable and confirming the transaction. Once the payment is successful, the system updates the inventory, generates the receipt, and completes the order. This diagram clearly shows the sequence of actions, decision points, and interactions between the user and the system, ensuring smooth and error-free store operations.



BLACKBOX TESTING:

Black Box Testing in the Departmental Store Management System checks system functionality by validating inputs (product, quantity, payment) against expected outputs (bill, receipt, stock update) without examining internal code.

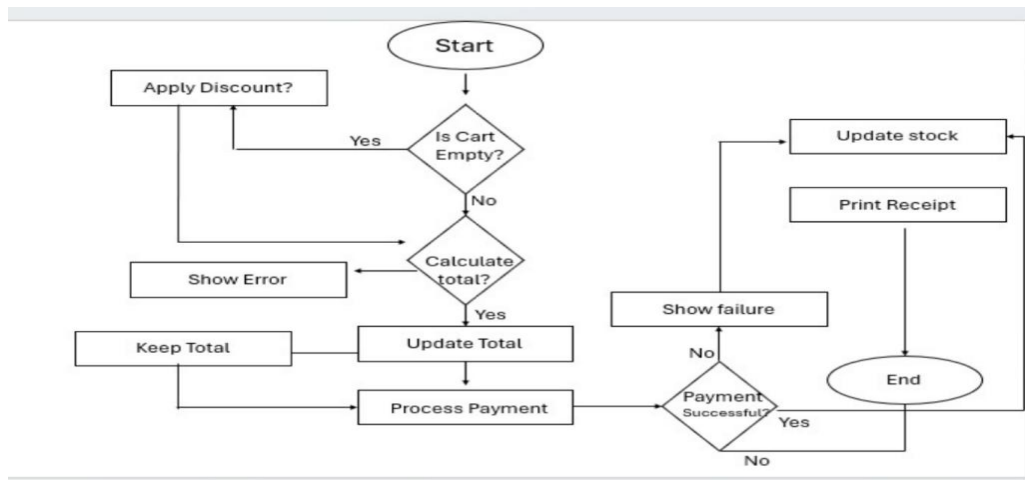
- ✦ Login function testing
- ✦ Product search testing
- ✦ Stock Availability testing
- ✦ Add to cart testing
- ✦ Payment process testing
- ✦ Receipt & sales record testing



WHITEBOX TESTING:

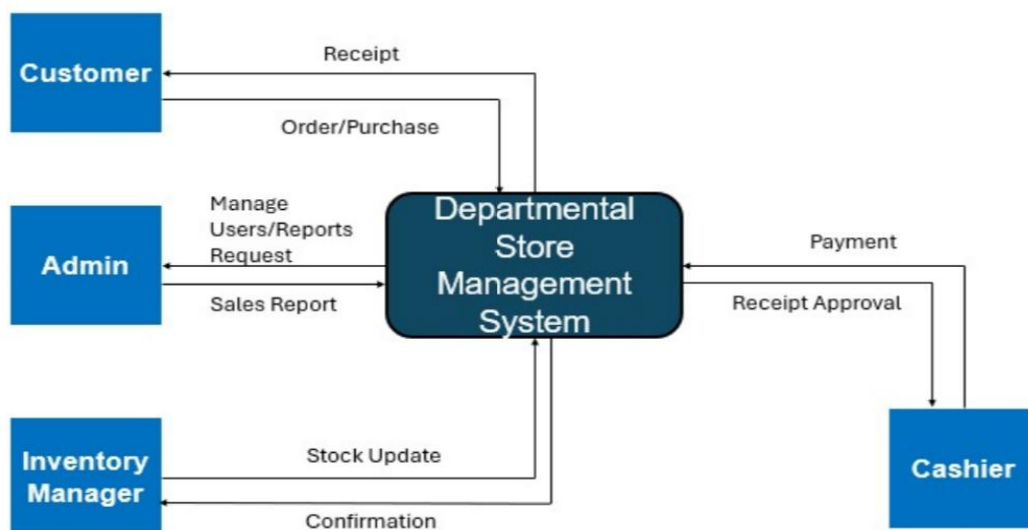
White Box Testing in the Departmental Store Management System focuses on testing the internal logic and control flow of modules such as cart handling, payment processing, discount application, and stock updates to ensure all decision paths and conditions execute correctly.

- ✦ Login module internal testing
- ✦ Product search login testing
- ✦ Stock checking code testing
- ✦ Add to cart function testing
- ✦ Payment processing internal testing
- ✦ Receipt & sales logging code testing



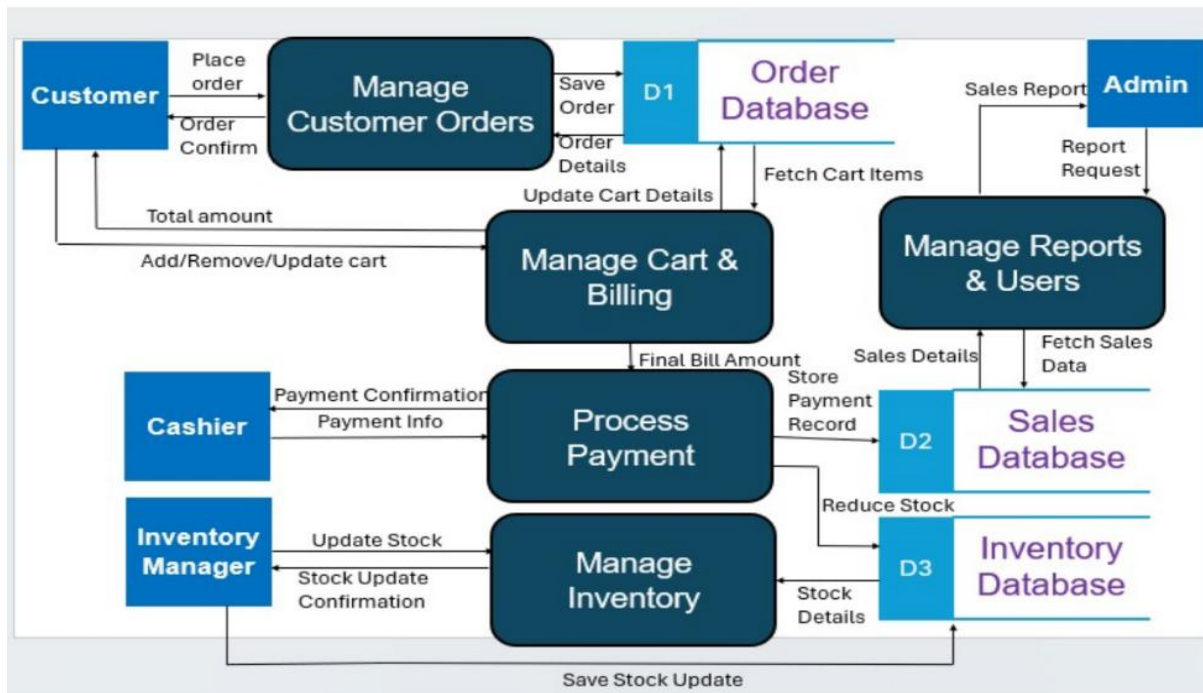
DFD LEVEL 0:

DFD Level 0 represents the overall data flow of the Departmental Store Management System as a single process. It shows how external entities such as Customer, Cashier, Admin, and Inventory Manager interact with the system. The system processes orders, payments, inventory updates, and report generation while storing and retrieving data from relevant databases.



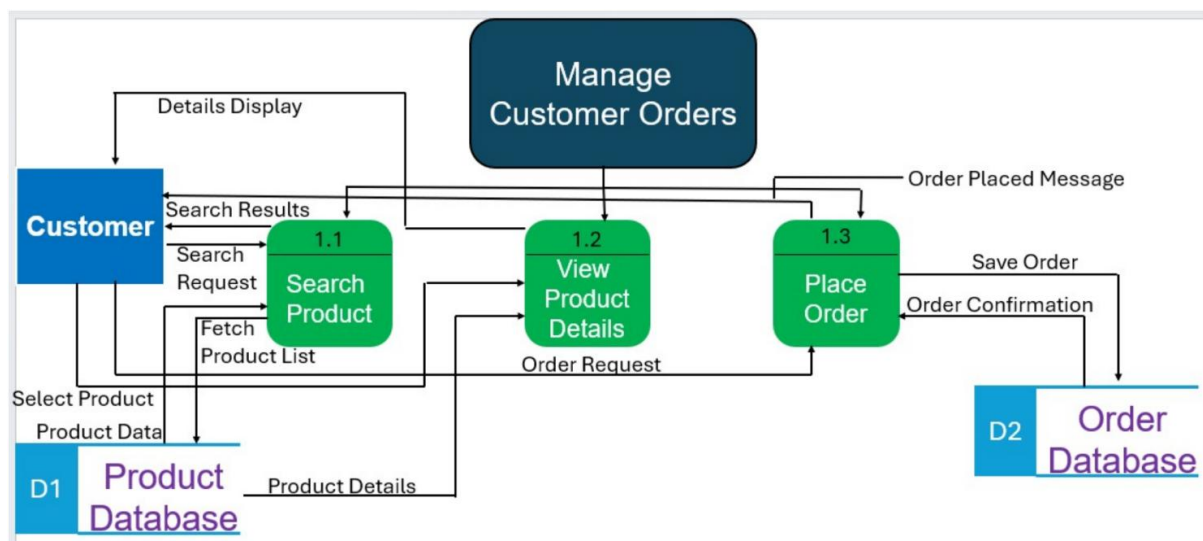
DFD LEVEL 1:

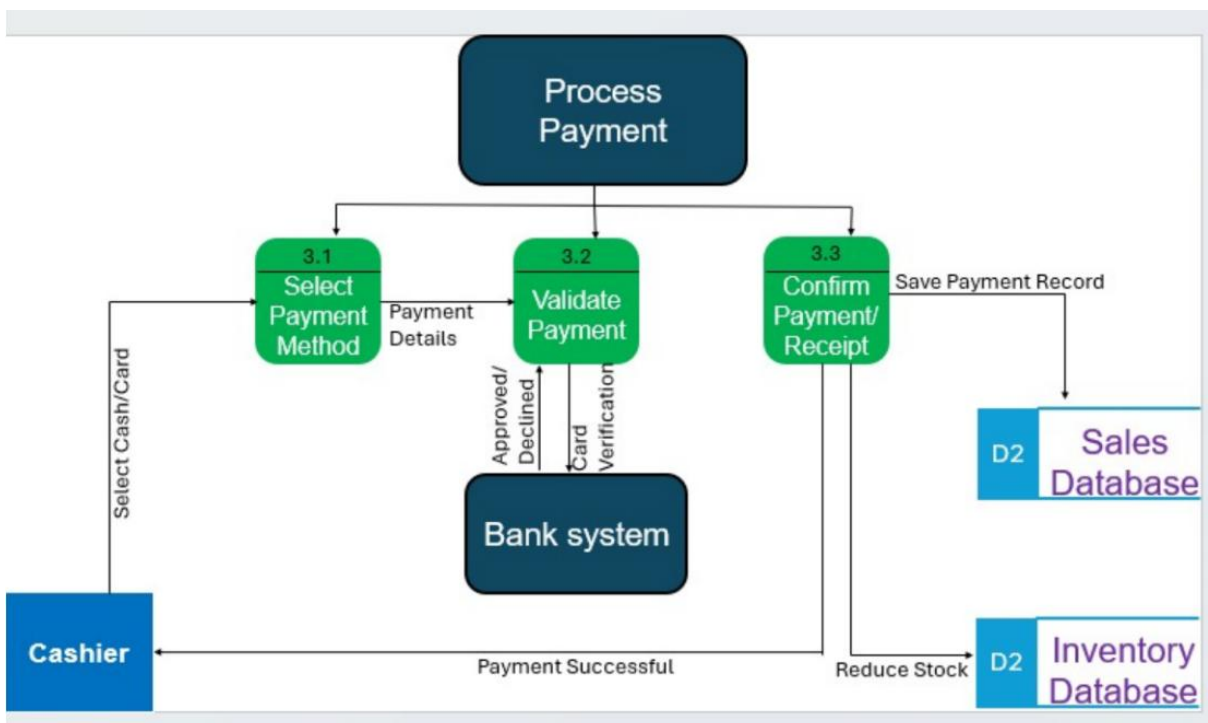
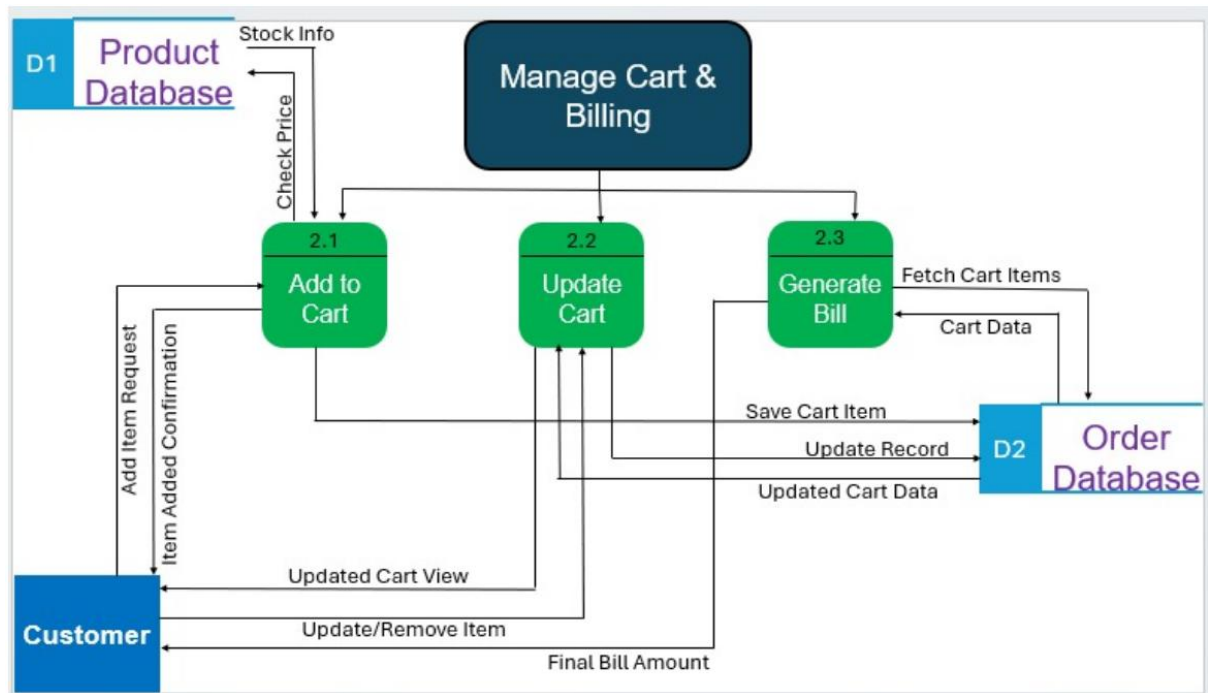
DFD Level 1 of the Departmental Store Management System divides the system into processes like order processing, payment handling, inventory update, and report generation, showing data flow between users and databases.

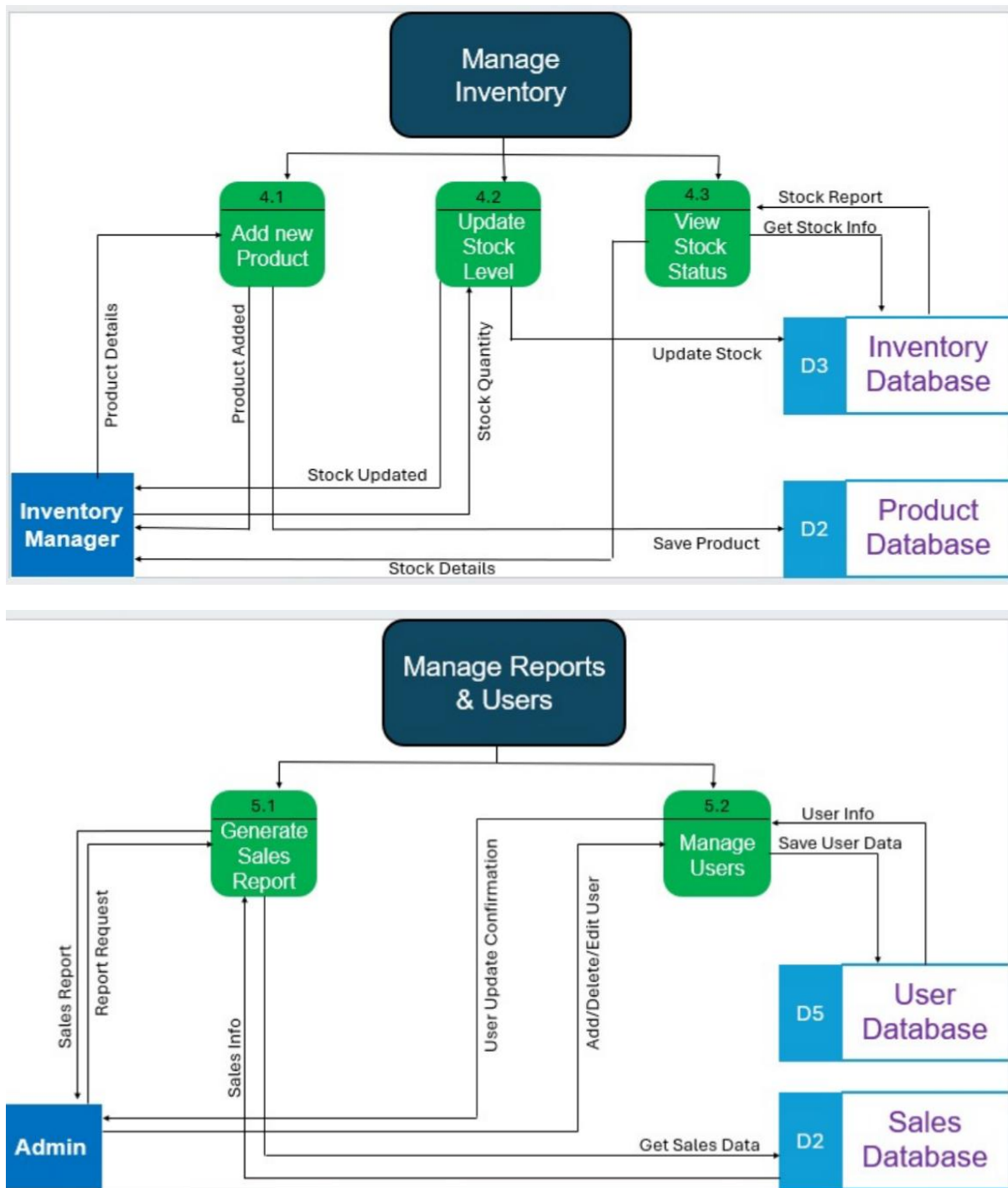


DFD LEVEL 2:

DFD Level 2 of the Departmental Store Management System further decomposes a Level 1 process (such as Order Processing) into detailed steps like product selection, cart update, discount application, payment verification, stock update, and receipt generation, showing precise data flow between these sub-processes and related databases.







PROCESS MODEL:

In Incremental Model, The system is developed in small modules (increments). Each module is built and tested individually, then integrated to form the complete system.

Project Modules:

- ✦ **Inventory Management:** Add/update/delete products, track stock
- ✦ **Billing & Sales:** Automatic bill generation, daily sales tracking

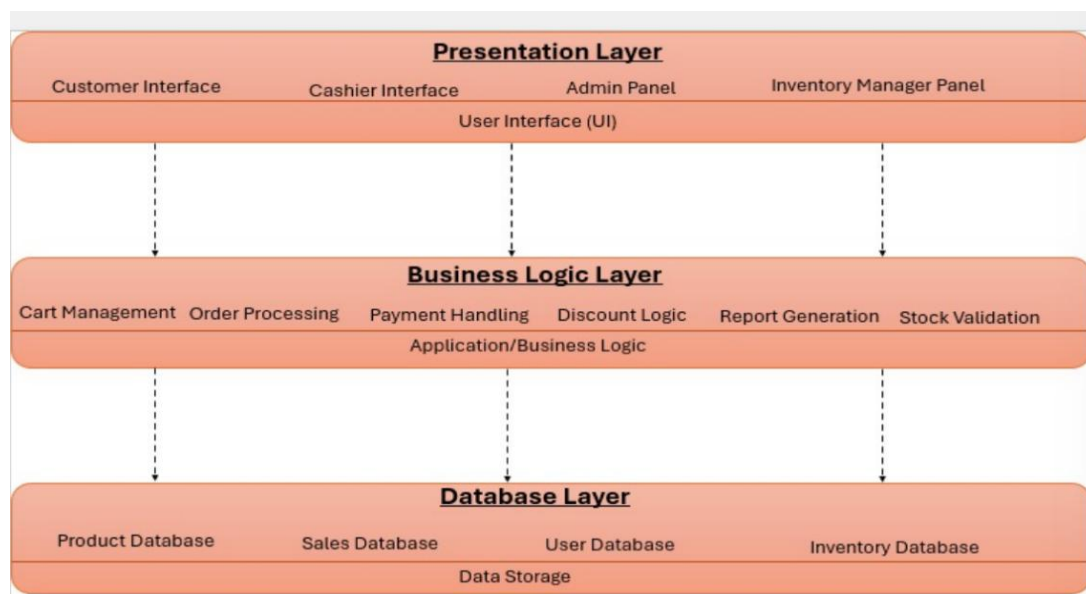
- ✦ **Customer & Supplier Management:** Maintain customer and supplier info
- ✦ **Reports & Analytics:** Generate daily/monthly sales reports, low stock alerts.

ARCHITECTURAL DIAGRAM:

Architectural Diagram for the Departmental Store Management System follows a 3-Tier Layered Architecture:

- ✦ **Presentation Layer:** Customer, Cashier, Admin, and Inventory Manager interfaces.
- ✦ **Business Logic Layer:** Cart management, order processing, payment handling, discount and report logic, stock validation.
- ✦ **Data Layer:** Product, Inventory, Sales, and User databases.

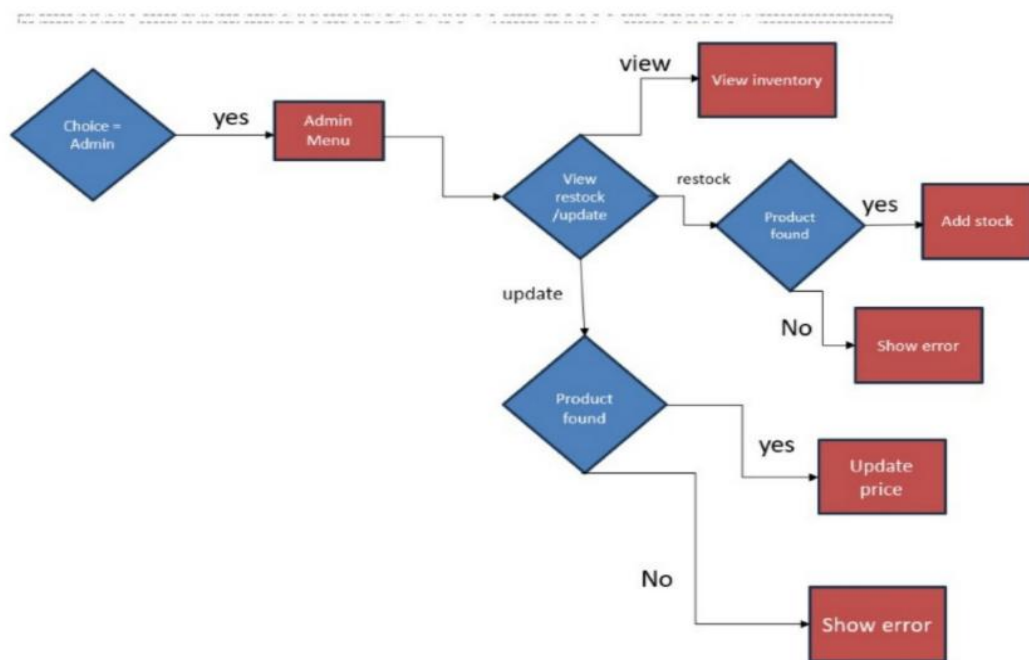
It shows clear separation of concerns, with all user requests passing through the business layer to interact with the data layer.



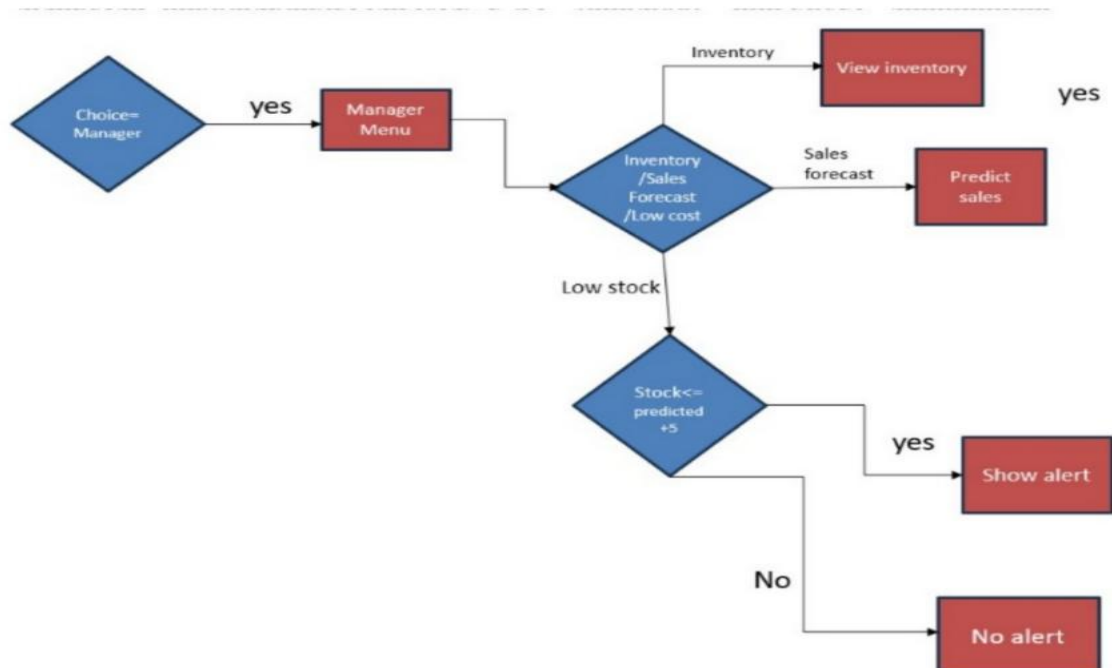
DECISION TREE:

This decision tree shows the step-by-step flow for handling an order, checking cart status, applying discounts, processing payment, handling errors, updating stock, and generating receipts. It captures all key points in the system.

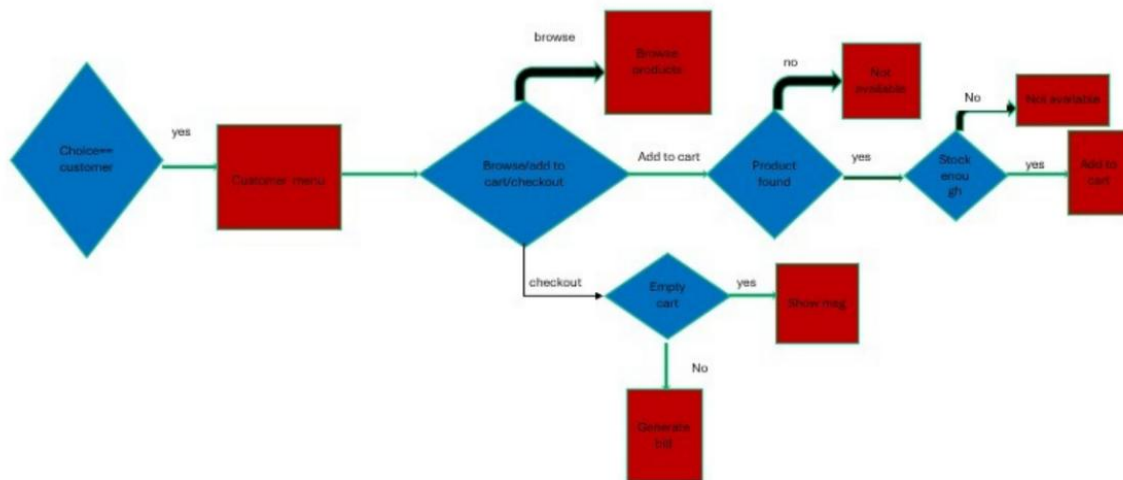
DECISION TREE FOR ADMIN:



DECISION TREE FOR MANAGER:



DECISION TREE FOR CUSTOMER:



DECISION TABLE:

It ensures that all scenarios—such as empty cart, invalid discount, or payment failures are handled correctly, helping testers verify system behavior systematically.

DECISION TABLE FOR MANAGER:

Conditions / Rules	R1	R2	R3	R4	R5
Choice = Manager	Y	Y	Y	Y	N
Sales selected	Y	N	N	N	-
Inventory selected	N	Y	N	N	-
Low stock	Y	N	-	-	-
Alert required	Y	N	-	-	-
-----	-----	-----	-----	-----	-----
Actions					
Manager Menu	X	X	X	X	-
Predict Sales	X	-	-	-	-
View Inventory	-	X	-	-	-
Show Alert	X	-	-	-	-
No Alert	-	X	-	-	-

DECISION TABLE FOR CUSTOMER:

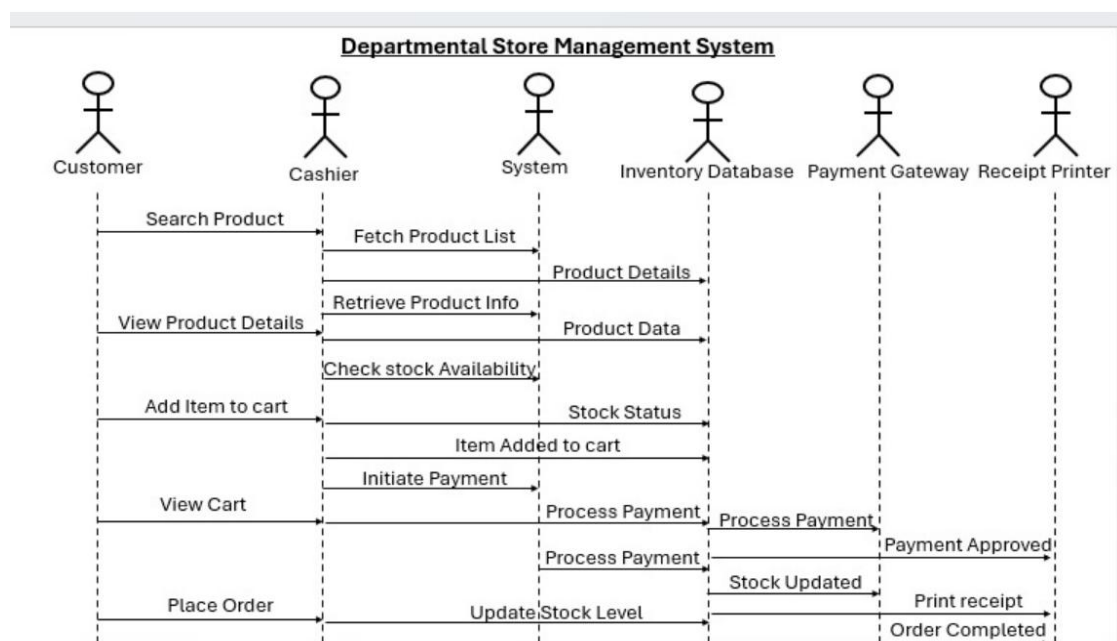
Conditions/Rules	R1	R2	R3	R4
Product found?	Y	Y	Y	N
Stock sufficient?	Y	Y	N	–
Cart empty at checkout?	N	Y	–	–
Checkout selected?	Y	Y	N	N
-----	-----	-----	-----	-----
Actions				
Add item to cart	X	X	–	–
Show recommendation	X	X	–	–
Show “Not available”	–	–	X	X
Generate bill	X	–	–	–
Calculate tax	X	–	–	–
Clear cart	X	–	–	–
Show “Thanks for shopping”	X	–	–	–

DECISION TABLE FOR ADMIN:

Conditions / Rules	R1	R2	R3	R4	R5	R6
Is Admin?	Y	Y	Y	Y	Y	N
View Inventory?	Y	N	N	N	N	–
Modify Inventory?	N	Y	Y	Y	N	–
Restock?	–	Y	N	–	–	–
Update Price?	–	–	–	Y	N	–
Actions						
Show Inventory	X	–	–	–	–	–
Add Stock	–	X	–	–	–	–
Update Price	–	–	–	X	–	–
Show Error	–	–	X	–	X	X

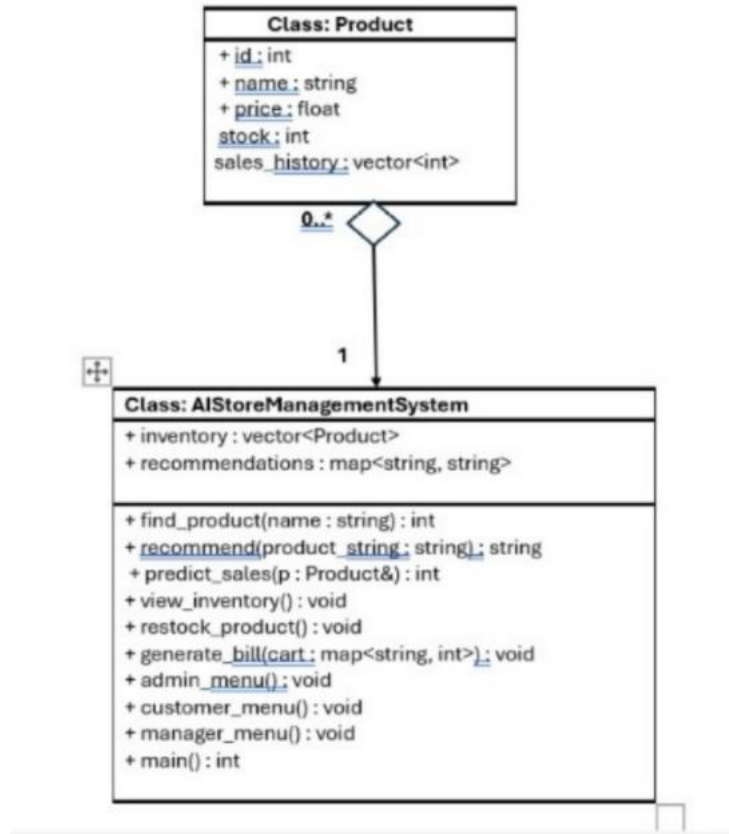
SEQUENCE DIAGRAM:

The sequence diagram shows the interaction between the Customer, Cashier, System, Inventory Database, Payment Gateway, and Receipt Printer. It illustrates the flow from searching products, adding items to the cart, and placing orders, to processing payment, updating stock, and generating receipts, capturing the step-by-step sequence of system operations.



CLASS DIAGRAM:

The class diagram models all key entities of the Departmental Store Management System, their attributes, methods, and relationships. It shows how Customers, Cashiers, Admins, and Inventory Managers interact with Products, Cart, and Orders to perform store operations efficiently.



THE END